ECEN/CSEN 123 Lab 6 Prelab
Dylan Thornburg, Austin Petersen, Madeline Follosco

Preparation:

• Read this lab.

• Review all of the documents posted in the parts folder related to the components in this lab and at the links in these documents (and the links from those links) so that you understand how the servo and the stepper motor work. Remember that I provided a component list at the beginning of the course.

• Figure out how to mount your servo and one of the distance sensors to your platform.

• Make sure you have your code from Lab 2.

Pre-Lab Report:

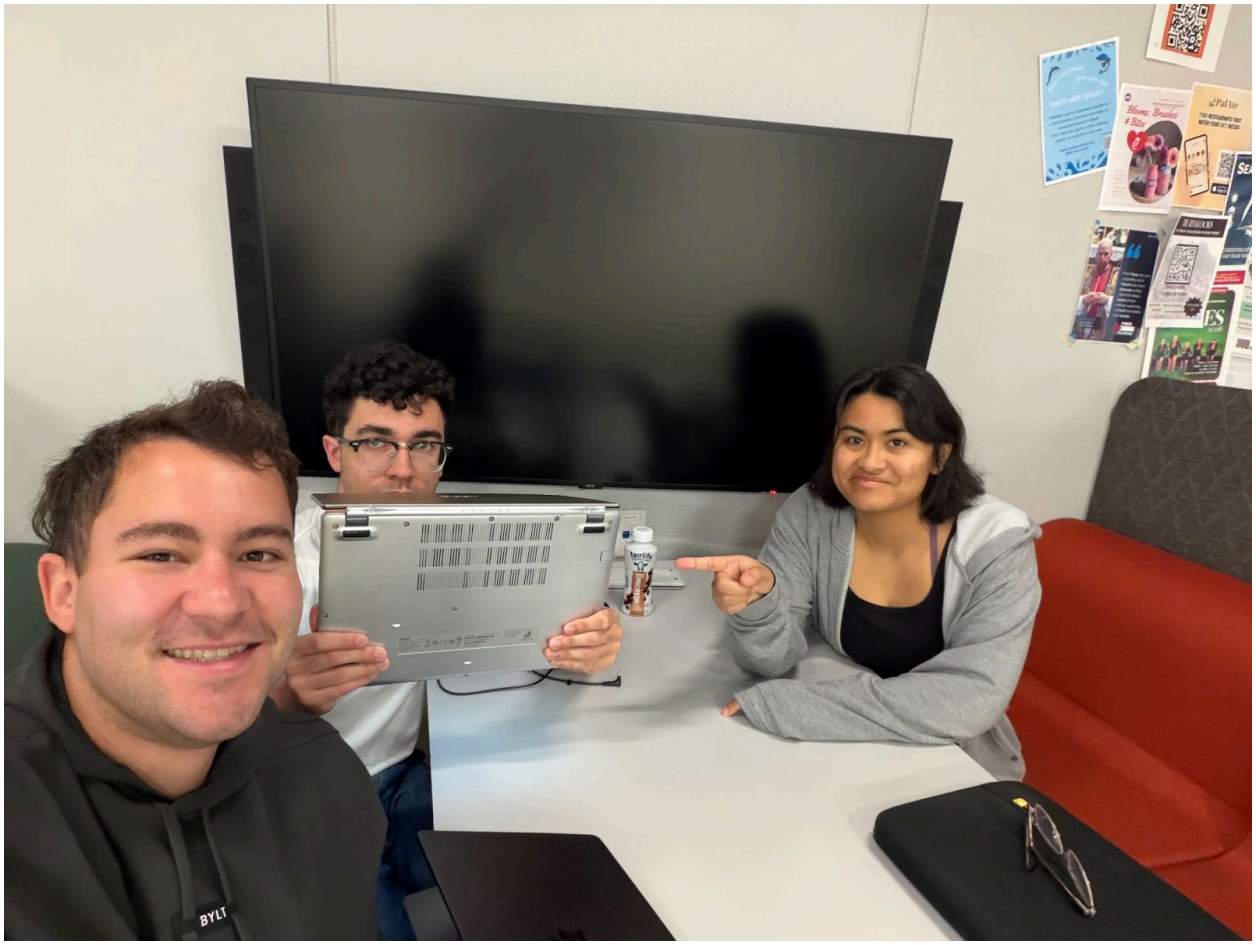
• Determine the maximum torque provided by the servo with a 4.8V power supply.
18.09 oz*in = 0.128Nm

• Determine the maximum torque provided by the stepper motor with a 12V power supply at 200steps/s.
100 gf*cm = .0098Nm

• Explain which motor has more torque.
The servo motor has more torque than the stepper.

• Peak torque in a 2026 BMW i4 M60 is 795 Nm. https://www.bmw-m.com/en/all-models/overview-m-and-m-performance/bmw-i4-m60-xdrive/2025/bmw-i4-m60-xdrive.html. How many lab stepper motors would it take to equal the torque in that BMW? (Pay attention to units).
795/.0098 = 81122.449
It would take 81,122.449 which is ~ 81,123 motors.

• Include a selfie from your planning meeting including something not from lab that contains an electric motor.



The laptop has fan motors.